Bound-State QED on S^3: The Leading One-Loop Hydrogen Lamb Shift

GeoVac Project

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Abstract

The hydrogen $2S_{1/2} - 2P_{1/2}$ Lamb shift at leading one-loop order, $\alpha(Z\alpha)^4$, assembled in the Eides–Grotch–Shelyuto § 3.2 convention with the Drake–Swainson tabulated Bethe logarithms, is 1052.19 MHz, against the experimental 1057.845 MHz (residual −5.65 MHz, −0.534%). This headline is textbook one-loop arithmetic with literature inputs; the GeoVac content is the framework's own Bethe logarithms, which reproduce the tabulated $2P$ and $3D$ values to 2.4% and 0.07%, while the $2S$ acceleration-form value is not yet converged (with them the total would be about 1055.6 MHz). The one-loop $\alpha(Z\alpha)^5$ term ($A_{50} = 4\pi(139/128 - \frac{1}{2}\ln 2) \approx 9.29$, about +6.9 MHz) is not included, so the residual is not a measure of missing multi-loop physics.

The Bethe-logarithm machinery came out of a six-sprint sequence (LS-1 through LS-6a, May 2026) that removed three structural obstructions. **LS-3** replaced the velocity-form Bethe-logarithm spectral sum, which diverges for $\ell > 0$ states because the closure $\sum_m |\langle 2P|p|m\rangle|^2(E_m - E_{2P}) \to 0$ vanishes exactly at $\ell > 0$ (the wavefunction at the origin vanishes), with the acceleration form for $2S$ (whose agreement with the tabulated value at $N = 40$, −0.92%, is a crossing: +2.98% at $N = 50$), leaving $2P$ structurally divergent. **LS-4** closed the $2P$ divergence by introducing the Drake–Swainson asymptotic-subtraction projection (Paper 34's 13th projection), producing K-independent regularized values for $\ell > 0$ Bethe logs through a two-piece spectral split with an explicit structural denominator $D_{\mathrm{drake}}(n,\ell) = 2(2\ell+1)Z^4/n^3$. **LS-6a** re-derived the original LS-1 one-loop self-energy in the canonical Eides–Grotch–Shelyuto § 3.2 convention and identified that LS-1 had inadvertently subtracted the Uehling kernel constant $4/15 = (10/9 - 38/45)$ from the canonical 10/9 self-energy coefficient, double-counting Uehling as a separate vacuum-polarization contribution. The convention fix is exactly the Uehling kernel, shifting the predicted Lamb shift by +27.13 MHz at $n = 2$, $Z = 1$.

Sprint LS-7 (May 2026) made a first pass at the two-loop self-energy on Dirac-S^3 via iterated CC spectral action: the pre-factor $(\alpha/\pi)^2(Z\alpha)^4/n^3$ of the dominant two-loop SE piece emerges from iterated proper-time integration, consistent with Paper 35's Prediction 1 at the structural level. An LS-7 decomposition of the residual into multi-loop and non-loop pieces is withdrawn: its rows do not sum to its stated totals and it omits the one-loop $\alpha(Z\alpha)^5$ term; the residual's attribution is open.

The framework's Bethe-logarithm machinery is built from projections that each have a literature precedent: the Coulomb Sturmian basis at exponent $\lambda = Z/n$ (the Fock label set reparameterized for bound-state space; Paper 7) and the Drake–Swainson asymptotic-subtraction projection (Paper 34 § III.13). No GeoVac-specific fitting parameter enters.

1 Introduction

The hydrogen Lamb shift is the canonical test of bound-state quantum electrodynamics. The $2S_{1/2} - 2P_{1/2}$ energy splitting at 1057.845 MHz, first measured by Lamb and Retherford in 1947, is predicted by QED to leading order through Bethe's 1947 non-relativistic calculation [13], refined

over decades to include relativistic, recoil, finite-nuclear-size, and multi-loop contributions [18]. At the time of writing the QED predicted value matches the measured value to better than parts in 10^7, requiring two-loop and three-loop self-energy contributions that have been the subject of dedicated computational efforts since the 1950s [19].

This paper documents the GeoVac framework's one-loop Bethe-logarithm machinery for the hydrogen Lamb shift. The headline number is the leading-order one-loop assembly with literature Bethe logarithms; the framework's own logarithms are compared with them.

Scope of the claim. -0.534% (-5.65 MHz) is roughly three orders of magnitude looser than the precision-QED frontier, which routinely reaches sub-kHz on the same observable with multi-loop self-energy and recoil corrections [18, 22]. The claim here is therefore not that the GeoVac framework competes with precision-QED compilations on the absolute number — it does not — and not that the 1052.19 MHz headline is a GeoVac computation: it is the textbook leading one-loop assembly with the Drake–Swainson tabulated Bethe logarithms, and it omits the one-loop $\alpha(Z\alpha)^5$ term. The GeoVac content is the Bethe-logarithm machinery (Sturmian basis at $\lambda = Z/n$, acceleration form, Drake–Swainson asymptotic subtraction), which reproduces the tabulated $2P$ and $3D$ logarithms and, for $2S$, has not yet converged.

The bound-state Lamb shift on Dirac-S^3 uses three standard projections in addition to the label set:

1. the **Fock conformal projection** (Paper 7), which maps the hydrogenic spectrum to the unit S^3 at focal length $p_0^2 = -2E$;
2. the **Camporesi–Higuchi spinor lift** (Paper 28 § IV; Paper 34 § III.7), which gives the Dirac-S^3 spectrum $|\lambda_n| = n + 3/2$ with degeneracy $g_n = 2(n+1)(n+2)$;
3. the **Coulomb Sturmian projection** at $\lambda = Z/n$ (Paper 8–9; LS-2 sprint), which the framework identifies as the Fock label set re-parameterized for bound-state spectral sums.

These three are required by the structure of the calculation. A fourth projection became necessary at LS-4 to close the $2P$ Bethe-log divergence: the **Drake–Swainson asymptotic-subtraction projection**, added as the 13th projection in Paper 34's dictionary. The fourth was a discovery, not a pre-existing piece of the framework; we discuss it in Section 3 below.

A fifth piece, the **Eides § 3.2 canonical one-loop SE convention**, is not a GeoVac projection in the Paper 34 sense — it is the standard textbook convention for organizing the one-loop self-energy contribution into Bethe-logarithm and finite-constant parts. Sprint LS-6a identified that the original LS-1 implementation used a different grouping of the constant terms that double-counted the Uehling kernel. We discuss this in Section 4 below.

Section 5 collects the final numerical result. Section 6 closes with the remaining open question: the LS-7 multi-loop test of Paper 35's Prediction 1.

Sprint provenance

The hydrogen Lamb chain is continuously verified by `tests/test_paper36_lamb_chain.py` with the Sturmian machinery ported durably to `tests/paper36_lamb_support/` (2026-07-03); the original sprint evidence is in `debug/ls{1,2,3,4,6a}_*.md` and the corresponding scripts and JSON data files. The framework infrastructure modules used are `geovac/qed_self_energy.py`, `geovac/qed_vacuum_polariza` `geovac/dirac_matrix_elements.py`, and the LS-3/LS-4 bound-state Sturmian / Drake–Swainson scripts under `debug/`.

2 Architecture: Dirac-S^3 plus Sturmian projection

The bound-state Lamb shift on Dirac-S^3 requires the spectrum, the bound-state wavefunctions, and the spectral sums for the self-energy and vacuum polarization.

2.1 The Dirac-S^3 spectrum

The Dirac operator on the unit S^3 has eigenvalues $\pm|\lambda_n| = \pm(n + 3/2)$ for $n = 0, 1, 2, \ldots$, with degeneracy $g_n = 2(n + 1)(n + 2)$ per chirality (Camporesi–Higuchi [23]). This is the bare spinor Laplace–Beltrami spectrum; it is π-free in $\mathbb{Q}$ at every n (Paper 35; the half-integer Hurwitz shift $3/2$ stays in $\mathbb{Q}$ because Bernoulli polynomials have rational coefficients). In the GeoVac *algebraic registry* (CLAUDE.md § 12), the Dirac matrix elements in the (κ, m_j) basis are algebraic-rational [2]: angular elements via Szmytkowski reductions, diagonal hydrogenic radial elements via Bethe–Salpeter rationals, off-diagonal radial elements via Hellmann–Feynman or Kramers–Pasternak direct integration. Nothing in this layer of the calculation introduces π.

2.2 The Coulomb Sturmian projection at $\lambda = Z/n$

The bound-state Lamb shift requires sums over excited bound-state spectra and continuum pseudostates. The Coulomb Sturmian basis at exponent λ is a discrete basis spanning the entire single-particle spectrum (bound plus continuum), with the special property that at $\lambda = Z/n$ for principal quantum number n, the basis contains the exact n-th hydrogenic eigenstate. Sprint LS-2 verified that the Sturmian basis at this exponent is, up to reparameterization, the Fock label set itself: the relabelling $(n, \ell, m) \to (\nu = n + n_{\text{bound}}, \ell, m)$ with appropriate Z/n-rescaling maps the discretized continuum pseudostates onto the same node lattice as the bound-state spectrum. This is the structural content of the LS-2 result quoted in Paper 34 § III.5: the "Sturmian projection" is not a separate projection beyond the Fock projection, but the same projection re-parameterized.

In bound-state spectral sums of the form

$$S(O) = \sum_m \frac{|\langle f|O|m\rangle|^2 (E_m - E_f)^p}{(E_m - E_f)^q} \tag{1}$$

for some operator O (the momentum p in the velocity form of the Bethe log; the acceleration $\hat{a} = (Z/r^2)\hat{r}$ in the acceleration form) and some powers p, q, the sum runs over all intermediate states m, including bound-state and continuum contributions. In the Sturmian basis at $\lambda = Z/n$ the sum reduces to a finite (truncated at N) sum over Sturmian basis indices, with the truncation error controlled by Schwartz / asymptotic expansions.

3 The Bethe logarithm: velocity, acceleration, Drake–Swainson

The Bethe logarithm for a hydrogenic state $|nl\rangle$ is defined as

$$\ln k_0(n, \ell) = \frac{\sum_m |\langle nl|\vec{p}|m\rangle|^2 (E_m - E_{nl}) \ln |E_m - E_{nl}|/\text{Ry}}{\sum_m |\langle nl|\vec{p}|m\rangle|^2 (E_m - E_{nl})}. \tag{2}$$

The denominator $I_v(nl) = \sum_m |\langle nl|\vec{p}|m\rangle|^2 (E_m - E_{nl})$ is the velocity-form sum rule; the numerator $J_v(nl)$ is the same sum weighted by $\ln |E_m - E_{nl}|/\text{Ry}$. For s-states the closure of I_v gives the standard contact-density result $I_v(nS) = (Z^4/n^3)$ (in atomic units, including the spin factor of 2) [21].

3.1 Velocity form: 2S works, 2P diverges

LS-2 implemented the velocity-form Bethe log via the Sturmian projection at $\lambda = Z/n$. At a Sturmian basis size N the truncated values converge to $\ln k_0(2S) = 2.726$ (Drake–Swainson 1990 reference value 2.81177; LS-2 residual -3.1% at $N = 50$) and $\ln k_0(1S) = 2.924$ (reference 2.98413; -2.0% at $N = 50$).

For $2P$, however, the velocity-form denominator $I_v(2P) = (Z^4/n^3)\,\delta_{\ell,0}$ vanishes exactly: the $\ell > 0$ wavefunction has $\psi_{2P}(0) = 0$, so the contact-density contribution is identically zero in any basis size. The finite-basis J_v/I_v ratio is $0/0$, with J_v being itself finite. This is a structural floor: no Sturmian basis size can resolve it. The standard literature treatment regularizes the ratio by asymptotic-subtraction techniques: the integral-representation technique originates with Schwartz [20] (for the helium ground state) and was extended to hydrogen p-states by Drake–Swainson [21].

3.2 Acceleration form for s-states

Sprint LS-3 implemented the acceleration form, replacing $\vec{p}$ in the matrix elements with the Coulomb acceleration $\vec{a} = (Z/r^2)\hat{r}$. The acceleration form denominator satisfies a different sum rule and is non-vanishing for s-states (it inherits $\sim Z^6\langle r^{-4}\rangle$). At $N = 40$:

- $\ln k_0(2S) = 2.786$, residual -0.92%, a crossing rather than a converged value: at $N = 50$ the residual is $+2.98\%$. (An earlier "3.3× tighter" than the velocity form compared values at different N; at equal N the two forms are comparable.)
- $\ln k_0(1S) = 3.002$, residual $+0.60\%$ — at $N = 20$, not $N = 40$: the 2026-07-02 recompute (`tests/test_paper36_lamb_chain.py`) showed the $1S$ acceleration-form value is a *crossing point* of a non-converging drift, not a converged value (by $N = 40$ it has drifted to 3.268, $+9.5\%$). The $2S$ value at $N = 40$ is the converged production number; the $1S$ channel is quoted at its crossing for reference only.

For $2P$, however, the acceleration form denominator also vanishes ($r^{-2}\langle\ldots\rangle$ does not give a nonzero closure-rule expression at $\ell > 0$). The structural floor remained.

3.3 The Drake–Swainson asymptotic-subtraction projection (LS-4)

LS-4 closed the $2P$ structural floor by importing the Drake–Swainson regularization scheme [21] and reading it as a new GeoVac projection (Paper 34 § III.13 added it as the 13th). The structural form is to split the divergent ratio at an intermediate cutoff K,

$$\ln k_0(n,\ell) \;=\; \frac{1}{D_{\mathrm{drake}}(n,\ell)}\Big[\beta_{\mathrm{low}}(N,K) \;+\; \beta_{\mathrm{high}}(K)\Big], \tag{3}$$

where β_{low} is the partial spectral sum up to energy K, β_{high} is the regularized asymptotic tail, and $\beta_{low}(K) + \beta_{high}(K)$ *is independent of K over a plateau of ∼3.6 orders of magnitude in K*, as verified at $K \in [0.05, 200]$. The structural denominator

$$D_{\mathrm{drake}}(n,\ell) \;=\; \frac{2(2\ell+1)Z^4}{n^3} \tag{4}$$

extends the s-state closure value $I_v(nS) = 2Z^4/n^3$ to $\ell > 0$ via the angular degeneracy factor $2\ell+1$, recovering the correct normalization for the regularized log.

At $N = 40$:

- $\ln k_0(2P) = -0.0307$, residual +2.40% vs Drake–Swainson −0.030017.
- $\ln k_0(3D) = -0.005236$, residual +0.07% vs −0.005232 (reference corrected 2026-07-03: an earlier printing quoted a mistranscribed −0.005249 against the same table).

The $3D$ result is the cleanest from-scratch Bethe-log result in the framework. The $2P$ result converges monotonically with N (+2.4% at $N = 40 \to +1.24\%$ at $N = 55$); it would close to $< 1\%$ at $N = 100$.

Remark 1 (The Drake denominator is combinatorial-rational)**.** *$D_{drake}(n, \ell) = (\text{spin multiplicity}) \times (2\ell+1) \times (Z^4/n^3)$ is a purely combinatorial quantity — spin times angular degeneracy times hydrogenic density. In Paper 34's transcendental taxonomy, the denominator sits in the combinatorial-rational tier, while the regularization mechanism itself sits in the flow tier. The Drake–Swainson projection is therefore an instance where the structural denominator is algebraic-rational (a Layer 1-type quantity) but the projection that* produces *the denominator from a divergent spectral ratio is in the calibration / flow sector. This is the same dual structure as the Sturmian projection (re-uses Fock label-set data; the projection itself is calibration-tier).*

Remark 2 (Drake–Swainson is structurally a sibling of the Schwartz integral form)**.** *Schwartz [20] regularized the 2P Bethe log via an integral form on the continuum spectrum. The Drake–Swainson form discretizes the same regularization onto the Sturmian pseudostate basis directly, simplifying the implementation. The two are mathematically equivalent on the Bethe-log answer; the LS-4 use is the discrete form because it composes naturally with the bound-state Sturmian projection of LS-2/LS-3.*

4 The Eides convention reconciliation (LS-6a)

After LS-4 the bound-state Bethe-log machinery was complete: s-states converged via acceleration form (LS-3), $\ell > 0$ states converged via Drake–Swainson regularization (LS-4). The combined $2S$ Lamb shift at LS-4's converged N was approximately 1029 MHz, leaving a residual −2.77% against experimental 1057.845 MHz. This residual was originally attributed to two sources: a small truncation $T \approx +3.5$ MHz from the $2S$ Bethe-log convergence and a large approximation-order $A \approx -32.8$ MHz, the latter assumed to be the missing α^5 multi-loop QED contribution.

4.1 Sprint LS-5: scoping the multi-loop test target

Sprint LS-5 (May 2026) scoped the LS-7 multi-loop calculation and discovered that the Eides–Grotch–Shelyuto 2001 Table 7.4 cumulative α^5 multi-loop contribution to the $2S$ Lamb shift is approximately +7.10 MHz, not +29 MHz. This led to the LS-5 *reframing* of the LS-3 residual: $\sim 80\%$ of the apparent A-tier residual was inconsistent with the multi-loop literature and must be a one-loop convention mismatch within the existing LS-1 implementation. The LS-1 § 2.2 footnote had explicitly flagged a +38/45 self-energy coefficient that disagreed with the textbook ~ 1078 MHz one-loop SE value by $\sim 4\%$, with the comment "a different grouping of the magnetic-moment / Darwin / Karplus–Klein terms would close this."

4.2 Sprint LS-6a: identifying the Uehling double-counting

Sprint LS-6a re-derived the LS-1 one-loop self-energy in the canonical Eides–Grotch–Shelyuto § 3.2 convention. The standard form for the $nS_{1/2}$ self-energy is

$$\Delta E_{\mathrm{SE}}(nS_{1/2}) = \frac{\alpha(Z\alpha)^4}{\pi n^3} m_e c^2 \Big[\tfrac{4}{3} \ln \frac{1}{(Z\alpha)^2} - \tfrac{4}{3} \ln \frac{k_0(n,0)}{\mathrm{Ry}} + \tfrac{10}{9} \Big]. \tag{5}$$

The constant +10/9 is the canonical combined one-loop SE finite constant for s-states; it incorporates Bethe's original 5/6, the Karplus–Klein–Darwin contact contribution, and the Schwinger anomalous magnetic moment at $j = 1/2$. In the Eides convention *there is no separate Schwinger AMM term to add* for s-states; it is already inside the 10/9.

The LS-1 implementation used +38/45 in place of +10/9. The constant difference is

$$\frac{10}{9} - \frac{38}{45} \;=\; \frac{50-38}{45} \;=\; \frac{12}{45} \;=\; \frac{4}{15}. \tag{6}$$

The constant 4/15 **is exactly the Uehling kernel constant** that appears in the one-loop vacuum polarization shift for s-states,

$$\Delta E_{\mathrm{VP}}(nS) = -\frac{4}{15\pi} \cdot \frac{\alpha^3 Z^4}{n^3}\,\mathrm{Ha}. \tag{7}$$

LS-1 had inadvertently subtracted the Uehling kernel constant from the canonical 10/9 one-loop SE coefficient, leaving 38/45. Since the Uehling shift is already counted as a separate VP contribution (it is a distinct one-loop diagram, not part of the SE), this is a double-counting error. Removing the Uehling subtraction restores the canonical 10/9 and shifts the predicted Lamb shift by

$$\Delta_{\text{LS-1}\to\text{LS-6a}} = +\frac{4}{15} \cdot \frac{\alpha^3 Z^4}{\pi n^3} \cdot (\text{Ha-to-MHz}) \;=\; +27.13 \text{ MHz at } n = 2, Z = 1. \tag{8}$$

Observation 1 (LS-6a: Uehling kernel double-counting)**.** *The LS-1 implementation of the one-loop hydrogen self-energy inadvertently subtracted the Uehling vacuum-polarization kernel constant* 4/15 *from the canonical Eides § 3.2 self-energy coefficient* 10/9*, double-counting Uehling as a separate vacuum-polarization contribution. Restoring the canonical* 10/9 *shifts the Lamb shift by exactly* $(4/15)(\alpha^3 Z^4/\pi n^3) = +27.13$ *MHz at* $n = 2, Z = 1$*, identified structurally rather than fitted. The LS-6a Lamb shift is* 1052.19 *MHz, residual* −5.65 *MHz (*−0.534%*) against experimental* 1057.845 *MHz.*

5 Result and component breakdown

5.1 Numerical result

Quantity	LS-1 (MHz)	LS-6a (MHz)	Shift (MHz)
SE $2S_{1/2}$ constant	+38/45	+10/9	+4/15
SE $2S_{1/2}$ total	+1039.31	+1066.44	+27.13
VP $2S_{1/2}$ (Uehling)	−27.13	−27.13	0
Total $2S_{1/2}$	+1012.18	+1039.31	+27.13
SE $2P_{1/2}$ (−1/6)	−12.88	−12.88	0
VP $2P_{1/2}$	0	0	0
Lamb shift	+1025.06	+1052.19	+27.13
Experimental	+1057.845		
Residual	−32.78 (−3.10%)	−5.65 (−0.534%)	

The LS-6a residual is a 5.8× improvement over LS-1 in absolute error. It is the leading-order $(\alpha(Z\alpha)^4)$ one-loop value with tabulated Bethe logarithms; the one-loop $\alpha(Z\alpha)^5$ term is not included (Section 5.2).

Continuous verification vs archived record. The chain above is continuously verified (added 2026-07-02): `tests/test_paper36_lamb_chain.py` recomputes the full component table closed-form from the Drake–Swainson tabulated Bethe logs (LS-1 1025.06, LS-6a 1052.19, shift +27.13, residual −5.65 MHz / −0.534%, and the exact identity $10/9 - 38/45 = 4/15$ in exact rational arithmetic), and (slow-marked) recomputes the from-scratch Sturmian Bethe logarithms of §3 at $N = 40$ ($2P = -0.0307$, $3D = -0.005236$, $2S$ acceleration = 2.786, plus the $1S$ $N = 20$ crossing and its $N = 40$ drift) from the machinery ported durably to `tests/paper36_lamb_support/`. The remaining §3 numbers — the LS-2 velocity-form values and the $N = 55$ $2P$ value -0.0304 ($|-1.24\%|$, re-validated 2026-07-02 at 62 s runtime) — stay archived-measured in the LS sprint record.

5.2 The remaining −5.65 MHz

The residual is not attributed here. The one-loop assembly above is the leading $\alpha(Z\alpha)^4$ order only; it omits the one-loop $\alpha(Z\alpha)^5$ term, $A_{50} = 4\pi(139/128 - \frac{1}{2}\ln 2) \approx 9.29$, which contributes about +6.9 MHz to the $2S$ level and by itself exceeds the residual, besides recoil, finite nuclear size and multi-loop terms of either sign. An earlier LS-7 decomposition of the residual (about +1.20 MHz multi-loop QED plus about +4.4 MHz of recoil, nuclear size and "hyperfine averaging") is withdrawn: its rows sum to +4.98 MHz and its non-loop rows to +3.78 MHz, neither matching the totals it stated; it did not include A_{50}; and the measured 1057.845 MHz interval is already defined between hyperfine centroids, so a +5 MHz "hyperfine averaging" row has no source. An order-by-order budget from Eides–Grotch–Shelyuto is owed before any attribution.

5.3 Sprint LS-7: structural confirmation of two-loop projection

Sprint LS-7 (May 2026) made a first pass at the two-loop self-energy contribution via iterated CC spectral action on the Dirac-S^3 spectrum. The structural finding: the two-loop SE prefactor

$$\Delta E^{(2L)}_{\mathrm{SE}}(2S) \;\sim\; \frac{(\alpha/\pi)^2(Z\alpha)^4}{n^3}\, m_e c^2 \cdot C_{2S}, \tag{9}$$

emerges naturally with C_{2S} a dimensionless bracket coefficient. The factor of $1/\pi^2$ in the prefactor traces to *two iterated Schwinger proper-time integrations*, exactly as Paper 35's Prediction 1 specifies for two-loop content (one π per loop, two loops, π^2 total). This is a structural confirmation of the prediction, but a weak test: the same $1/\pi^2$ prefactor appears in flat-space two-loop QED computations and does not distinguish GeoVac. The strong test is the native derivation of the bracket coefficient C_{2S} from the GeoVac bound-state Sturmian projection, which is the LS-8a sprint scope.

5.4 What π-content the calculation introduces

The Lamb shift contains explicit π in the standard QED forms (the $1/(\pi n^3)$ prefactor of the Eides expression (5); the $1/(48\pi^2)$ vacuum-polarization coefficient Π [4]; the $4/(15\pi)$ Uehling shift, etc.). Under Paper 35's classification, every π in the calculation traces to exactly one source: the continuous Schwinger proper-time integration at one loop. This is the calibration-$\pi^{2k}\cdot\mathbb{Q}$ tier of Paper 18 and the "observation / temporal-window projection" of Paper 34 § III.15. No π appears in the bare Dirac-S^3 spectrum (which is half-integer rational, π-free in $\mathbb{Q}$); no π appears in the bare Sturmian basis at $\lambda = Z/n$ (which re-uses the same Fock label set, π-free); no π appears in the Drake–Swainson denominator $D_{\mathrm{drake}} = 2(2\ell+1)Z^4/n^3$ (combinatorial-rational); π appears via the Schwinger proper-time integral that produces the $\Pi = 1/(48\pi^2)$ coefficient and the Eides $1/(\pi n^3)$

prefactor. Both of these are Paper 35's "observation projection" — a continuous integration over a temporal/spectral parameter promoted from the discrete graph.

The Lamb shift therefore enters Paper 35's empirical confirmatory list: every π in the calculation has the predicted Paper 34 provenance.

6 Outlook: LS-7 and the multi-loop test

The remaining -5.65 MHz residual is not attributed (Section 5.2); an order-by-order budget, starting with the one-loop $\alpha(Z\alpha)^5$ term, is owed first. The follow-on sprints were planned along these lines:

- **Multi-loop QED sprints (LS-8a, LS-8b, LS-8c).** These are the contributions for which Paper 35's Prediction 1 in the multi-loop sector is directly testable via iterated CC spectral action.
- **Non-loop sprints (LS-8d, LS-8e).** Recoil and finite-nuclear-size corrections via the rest-mass projection (Paper 34 § III.14) and the nuclear-electronic composed Hamiltonian machinery of Paper 23. Hyperfine averaging via the $I \cdot S$ hyperfine coupling already implemented in Paper 23 (the $3A_{\mathrm{hf}}/4 = 1.62 \times 10^{-7}$ Ha singlet-triplet gap that reproduces the 21 cm line).

Proposition 1 (LS-7 multi-loop test, sharpened)**.** *The two-loop Connes–Chamseddine spectral action on the Dirac-S^3 spectrum, with bound-state Sturmian projection at $\lambda = Z/n$ and Drake–Swainson regularization on inner spectral sums for $\ell_{internal} > 0$, should reproduce the two-loop self-energy contribution to the 2S Lamb shift. The structural pre-factor $(\alpha/\pi)^2(Z\alpha)^4/n^3$ for the dominant two-loop SE piece was confirmed at LS-7; the strong test is the native derivation of the dimensionless bracket coefficient $C_{2S} = +3.63$ (current independent state-of-the-art two-loop SE computations at low Z are reported by Yerokhin, Harman, and Keitel [22]) from the bound-state Sturmian projection, which is the LS-8a sprint scope.*

LS-7 verified that the framework's two-loop machinery is structurally adequate: `geovac/two_loop_self_energy` (backed by `tests/test_two_loop_self_energy.py`) computes the two-loop SE bracket; the separate `geovac/qed_two_loop.py` computes connected double Dirac Dirichlet sums in closed form via half-integer Hurwitz, and the $1/\pi^2$ prefactor of the two-loop SE emerges from iterated Schwinger proper-time integration as Paper 35 predicts. No structural obstruction was encountered.

The falsification criterion is sharper than the original framing. If LS-8a's native derivation of C_{2S} from iterated CC spectral action plus bound-state Sturmian gives a value far from $+3.63$ at the right sign and order, then the framework's iterated temporal-compactification projection does not reproduce flat-space two-loop QED bracket structure, and Prediction 1 needs revision in the multi-loop sector. If C_{2S} comes out at the right value without external calibration, the framework reproduces α^5 multi-loop QED to the same precision as it reproduced one-loop. Either way the result is structurally informative.

6.1 Sprint LS-8a result: WEAK verdict, structural confirmation with renormalization gap

Sprint LS-8a (May 2026) executed the strong test. The bare iterated Connes–Chamseddine spectral action on the Dirac-S^3 spectrum was implemented for both two-loop self-energy topologies (rainbow and crossed), with full SO(4) channel-count vertex selection at all four internal vertices and bound-state Sturmian projection at $n_{\mathrm{ext}} = 1$ (the Camporesi–Higuchi level corresponding to hydrogen

$2S$). The dimensionless bracket extracted with the volume-factor normalization $1/(4\pi)^2$ on the two photon legs is

$$C_{2S}^{\mathrm{GeoVac}}(n_{\max}) = \frac{1}{(4\pi)^2} \cdot \left[\Sigma_{\mathrm{rainbow}}(n_{\max}) + \Sigma_{\mathrm{crossed}}(n_{\max})\right]. \tag{10}$$

The numerical result is given in Table 1. At $n_{\max} = 4$ the value $C_{2S}^{\mathrm{GeoVac}} = 4.82$ lies within $1.33\times$ of the literature target $+3.63$ at the correct sign and order, but the spectral sum does NOT converge as $n_{\max} \to \infty$. A power-law fit gives raw total $\sim 6.6 \cdot N^{3.43}$ — the bare framework faithfully reproduces the flat-space two-loop QED UV divergence rather than producing a finite answer.

Table 1: LS-8a native two-loop SE bracket convergence. The bare iterated CC spectral action is UV-divergent; finite extraction requires renormalization counterterms (Z_2, δm) NOT present in the bare spectral action.

$n_{\max}$	raw total	C_{2S}^{GeoVac}	predicted MHz
2	44.82	0.284	0.067
3	266.3	1.687	0.399
4	761.3	4.821	1.139
5	1657.8	10.498	2.481
6	3052.8	19.332	4.568

Two natural regularization paths within the GeoVac toolkit fail to tame the divergence: (i) subtracting the disconnected $[\Sigma_{1L}]^2$ piece removes only $\lesssim 0.1\%$ of the total at every tested $n_{\max}$, confirming the divergence lives in the genuine connected two-loop diagram; (ii) Drake–Swainson asymptotic subtraction (Paper 34's 13th projection, used in LS-4 to close the $\ell > 0$ Bethe log) does not apply because the divergence is power-law ($\sim N^{3.43}$), not logarithmic. Finite extraction therefore requires field-theoretic renormalization counterterms (Z_2 wave-function renormalization, δm mass renormalization) implemented at the spectral level — these are NOT generated by the bare Connes–Chamseddine spectral action and would be a substantial multi-sprint extension to derive natively.

Proposition 2 (LS-7 multi-loop test, finalized after LS-8a)**.** *The bare iterated Connes–Chamseddine spectral action on Dirac-S^3 with bound-state Sturmian projection at $\lambda = Z/n$ correctly reproduces the structural prefactor $(\alpha/\pi)^2(Z\alpha)^4/n^3$ of the two-loop self-energy contribution to the $2S$ Lamb shift (LS-7) and faithfully inherits the UV-divergent integrand structure of flat-space two-loop QED (LS-8a). It does NOT autonomously extract the finite physical bracket $C_{2S} = +3.63$ because that extraction requires field-theoretic renormalization counterterms (Z_2, δm) NOT generated by the bare spectral action. Paper 35 Prediction 1 is consistent at the structural level (the π content of the prefactor traces to two iterated photon proper-time integrations) but is not strongly tested at two loops within the bare framework. The multi-loop QED contribution to the H $2S$ Lamb shift cannot be derived natively from GeoVac without an LS-8a-renorm sprint implementing Z_2 and δm at the spectral level.*

This is honest scope. GeoVac-internal projections (Paper 34's twenty-eight) handle the *finite* parts of one-loop bound-state QED via Drake–Swainson + Eides § 3.2 conventions (the leading one-loop order; the $\alpha(Z\alpha)^5$ term is not included). They do NOT autonomously regularize the multi-loop UV divergences; renormalization counterterms remain external input. The structural reading is that GeoVac controls the divergence *structure* of multi-loop QED faithfully but does not generate the renormalization counterterms that produce finite multi-loop predictions. This places

a clean scope boundary on the framework between the finite one-loop Bethe-logarithm sector (in reach) and multi-loop QED (requires renormalization machinery as separate input).

6.2 Sprint MH Track A: Muonic Hydrogen via the Rest-Mass Projection

The leading one-loop assembly transfers to muonic hydrogen (μp) under the rest-mass projection (Paper 34 § III.14), which swaps $m_e \to m_\mu = 206.7682830\, m_e$ in the lepton register. The CREMA 2010 measurement [Antognini et al. 2013] provides a precision- frontier benchmark: $\Delta E^{\mu p}_{\rm Lamb} = 202.3706(23)$ meV.

Architecture transfer. The Eides § 3.2 convention (LS-6a) and Drake–Swainson regularization (LS-4) carry over verbatim with the energy unit $\mathrm{Ha}_{\rm lepton} = \alpha^2 m_{\rm red} c^2$, where $m_{{\rm red},\mu p} = 185.84\, m_e$ (vs $0.999456\, m_e$ for ep). Bethe-log values $\ln k_0(2S) = 2.812$, $\ln k_0(2P) = -0.030$ are universal in dimensionless atomic units (properties of the hydrogenic Coulomb wavefunction, lepton-mass-independent).

The full Uehling kernel: architecturally new. Paper 36's normal-H closure uses a contact-density Uehling formula valid when $a_{\rm lepton} \gg \lambda_{C,e}$. For muonic hydrogen, $a_\mu \approx 285$ fm and $\lambda_{C,e} \approx 386$ fm overlap directly: the dimensionless Uehling parameter $\beta = 2 m_e a_\mu = 1.475$ (vs 274 for normal H) puts the calculation outside the contact regime, and naive contact-form scaling overshoots the muonic Uehling by $\sim 3.55\times$. The framework's response is full numerical integration of the Uehling kernel

$$U(s) = \int_1^\infty dt\; e^{-st} \left(1 + \frac{1}{2t^2}\right) \frac{\sqrt{1 - 1/t^2}}{t} \tag{11}$$

against the muonic $2S$ and $2P$ wavefunctions, yielding $\Delta E_U(2S) = -210.59$ meV, $\Delta E_U(2P) = -5.59$ meV, and $\Delta E^{\mu p}_{\rm VP} = +205.007$ meV. **Match to Antognini 2013 [14]:** < 1 **ppm** against the canonical literature value 205.0074 meV.

Numerical result. The component decomposition (Sprint MH Track A, 2026-05-08):

Component	Framework (meV)	Antognini 2013 (meV)
Uehling (full kernel)	+205.0074	+205.0074
Self-energy (muon, leading $m_{\rm red}$)	−0.83	−0.668
Friar moment ($r_p = 0.8409$ fm)	−3.675	−3.842
Framework subtotal	+200.50	—
Källén–Sabry 2-loop VP (lit. input)	—	+1.508
Higher-order multi-loop (lit. input)	—	+0.189
Recoil NLO (lit. input)	—	−0.045
Nuclear polarizability (lit. input)	—	+0.013
Framework + literature	+202.17	+202.16
Experimental (CREMA)	+202.3706(23)	
Residual	−0.10%	

Honest scope. The framework natively reproduces the dominant Uehling at < 1 ppm and the leading Friar moment at 4.3%. The self-energy gap (24% of value, 0.16 meV absolute) is leading-order $m_{\rm red}$ scaling, omitting next-to-leading $\alpha(Z\alpha)^4 \times (m_{\rm red}/m_p)$ recoil-mixing terms. The +1.67

meV literature-input total covers exactly the contributions blocked by the LS-8a renormalization wall (multi-loop QED δm, Z_2 counterterms not generated natively by the bare spectral action) plus QCD-internal nuclear polarizability (structurally outside the spectral-action framework). The proton radius puzzle is resolved (PDG 2024 [17]), so this sprint is a *framework calibration check* on the precision frontier of bound-state QED, not a new physics measurement.

Structural reading. Sprint MH Track A is the cleanest single-sprint demonstration of the multi-focal-composition machinery on a precision physics observable. Three takeaways: (i) the rest-mass projection works at the architectural level (clean transfer); (ii) the multi-focal architecture handles the muonic regime where the Bohr scale and electron Compton scale overlap, via the full Uehling integral; (iii) the leading one-loop assembly transfers under the rest-mass projection, with a sub-percent residual that is not attributed.

Sprint provenance: the driver is preserved durably at `tests/paper36_precision_support/sprint_mh_track_a` (resurrected 2026-07-04 from the pruned `debug/sprint_mh_track_a.py`; archived-measured, not yet regression-gated — re-validation is a named follow-up), with the original sprint memo and JSON in the `debug/` clean-room record.

6.3 Sprint Mu Lamb: Muonium Lamb Shift at Leading One-Loop Order

The leading one-loop assembly also transfers to muonium (Mu = $e^-\mu^+$) under the rest-mass projection.

Architecture transfer. Muonium has electron-as-lepton and antimuon-as-nucleus: $m_{\rm red}(e\mu^+) = 0.995169\, m_e$, ratio to ep is 0.995729. The Uehling parameter $\beta = 2m_e a_{\rm lepton} = 2/(m_{\rm red}\alpha) = 275.40$ vs 274.22 for normal H — essentially identical contact regime, so the Paper 36 LS-1..LS-7 architecture transfers verbatim with $m_{\rm red}$ rescaling of the energy unit. The full Uehling kernel and the contact form agree to ~ 0.24 MHz (0.9% of the Uehling contribution), confirming the contact regime.

Numerical result. Framework-native one-loop QED at $m_{\rm red}(e\mu^+)$:

Component	Framework (MHz)
SE($2S$) Eides § 3.2	+1061.31
SE($2P$) Eides § 3.2	−12.82
$\text{Lamb}_{\rm SE}$	+1074.13
VP full Uehling Lamb	−26.50
Framework total	+1047.63

An earlier version compared this total with a muonium Lamb theory reference attributed to Karshenboim 2005 (+1047.49(5) MHz, with component values and a recoil itemization); those figures could not be located in that source, so the comparison, the "+0.013%" residual and the recoil-absorption reading built on it are withdrawn until a verifiable reference is supplied.

Pure rest-mass rescaling cross-check. A trivial multiplication of the Paper 36 normal-H framework value 1052.19 MHz by the reduced-mass ratio 0.995729 gives 1047.696 MHz. The native Eides-bracket compute gives 1047.625 MHz. Agreement to within 0.071 MHz / 68 ppm of the value (corrected 2026-07-03: an earlier printing misstated this as 7 ppm). This is the sharpest verification yet of the rest-mass projection theorem (Paper 34 § III.14) at the Lamb-shift bracket level.

Mass-hierarchy matrix completion.

System	Mass hierarchy	Native res
H $2S$-$2P$ Lamb (Paper 36)	$m_{\rm red} \approx 1$, $m_n \approx 1836$	-0.534%
μp $2S$-$2P$ Lamb (Sprint MH-A)	$m_{\rm red} \approx 186$, $m_n \approx 1836$	-0.92%
Mu $1S$-$2S$ (Track 1)	$m_{\rm red} \approx 1$, $m_n \approx 207$	-0.11 ppr
Mu $2S$-$2P$ Lamb (this)	$m_{\rm red} \approx 1$, $m_n \approx 207$	(reference

Structural reading. Across the mass-hierarchy points the leading one-loop assembly transfers under the rest-mass projection; the residuals are not attributed term by term. Sprint provenance: the driver is preserved durably at `tests/paper36_precision_support/precision_catalogue_muonium_lamb.py` (resurrected 2026-07-04; archived-measured, not yet regression-gated), with the memo and JSON in the `debug/` record.

6.4 Muonium catalogue closure: triple completes at the LS-8a wall

The muonium catalogue closes with the third QED-channel test: the ground-state hyperfine splitting $\nu_{\rm HFS}$(Mu). Together with Track 1 (1S–2S, -0.11 ppm vs Mu-MASS) and the $2S$–$2P$ Lamb shift (§6.3, whose theory reference is withdrawn), this completes a three-channel test of the framework under the rest-mass projection (lepton unchanged, nucleus $m_p \to m_\mu$) with point-leptonic "nucleus" (no QCD budget):

Channel	Framework value	Residual
1S–2S transition	2,455,528,721 MHz (rest-mass rescaling)	-0.11 ppm
1S–2S transition	2,455,528,394 MHz (BF + SE + recoil)	-0.25 ppm
$2S$–$2P$ Lamb shift	1047.63 MHz (one-loop QED full Uehling)	-0.6% vs experiment (Paper 34)
1S hyperfine	4464.1905 MHz (BF + Schwinger one-loop)	+199 ppm

The 1S HFS computation uses the Sprint MH Track B Bohr–Fermi machinery applied to the leptonic-nucleus regime:

$$A_{\rm hf}^{\rm Mu} = \tfrac{2}{3}\, g_e\, g_\mu\, \alpha^2\, Z^3\, m_{\rm red}^3/(m_e\, m_\mu) \quad \text{[Hartree]} \tag{12}$$

with $m_{\rm red}(e\,\mu^+) = m_\mu/(m_e + m_\mu) \approx 0.99519\, m_e$, $Z = 1$ (single antimuon charge), and g_e, g_μ either Dirac = 2 or full CODATA values [16] (the difference being $a_e + a_\mu$ absorbed into the magnetic moment). Schwinger one-loop applies as multiplicative $(1 + \alpha/2\pi)^2$ acting on the strict-Dirac BF. No magnetization-density operator (point lepton); no cross-register recoil at this leading order (the natural Roothaan regime $\lambda_{\rm lepton} \ll \lambda_{\rm nucleus}$ is preserved at $1 \ll 207$, but the leading recoil is already captured inside $m_{\rm red}^3$).

The cleanest LS-8a isolation in the catalogue. Mu HFS is the only catalogue test whose Layer-2 input reduces to LS-8a multi-loop QED + recoil NLO alone, without any QCD competing budget. Hydrogen 21 cm has Zemach + multi-loop; muonic hydrogen HFS has electron-VP-in-muonic-potential; deuterium HFS has nuclear polarizability. Mu HFS has *none of these*. The +199 ppm framework residual (BF + Schwinger one-loop, no Layer-2 input) is not attributed: the omitted corrections include the leading recoil term, which is negative and of comparable size, as well as higher-order QED. An itemization of the missing corrections attributed to Karshenboim 2005, with all terms positive and said to close agreement to < 1 ppm, is withdrawn: adding positive terms to a positive residual cannot close it.

Mass-hierarchy ⊗ Layer-2-content matrix. Combining Mu HFS with the existing catalogue rows establishes a two-axis matrix. Each row is a mass-hierarchy regime (the rest-mass projection's input); each column is a Layer-2-content pattern (what the framework cannot autonomously generate):

	QCD-free (point/lepton nucleus)	**QCD content** (proton/deuteron nucl
Equal mass ($\lambda_a = \lambda_b$)	Ps HFS	—
Light ≪ heavy ($m_e \ll m_p$)	**Mu HFS** (this)	H 21cm, D HFS
Heavy ∼ heavy (overlap regime)	—	μH HFS, μH Lamb
Heavy in light bath	—	μH Lamb (reverse)

The Mu HFS row is structurally novel: it is the only QCD-free entry in the dominant $m_e \ll m_p$ class, and therefore the only system where the framework's leading-order isolation of the LS-8a wall is quantitatively unambiguous.

Sprint provenance. The driver is preserved durably at `tests/paper36_precision_support/precision_catalo` (resurrected 2026-07-04; archived-measured, not yet regression-gated), with the memo and JSON in the `debug/` record.

7 Conclusion

The leading one-loop assembly of the hydrogen Lamb shift, with literature Bethe logarithms, gives 1052.19 MHz (−0.534%); the GeoVac content is the Bethe-logarithm machinery built in a six-sprint sequence (LS-1 through LS-6a), whose three pieces are: (i) the acceleration-form Bethe logarithm in the Sturmian basis at $\lambda = Z/n$ (LS-3; for $2S$ not yet converged, its −0.92% at $N = 40$ being a crossing); (ii) the Drake–Swainson asymptotic-subtraction projection that closes the $\ell > 0$ structural floor (LS-4, Paper 34's 13th projection, with structural denominator $D_{\text{drake}} = 2(2\ell + 1)Z^4/n^3$ in the combinatorial-rational tier); and (iii) the Eides § 3.2 canonical convention for the one-loop self-energy coefficient (LS-6a, identifying that the LS-1 implementation had inadvertently subtracted the Uehling kernel constant 4/15 from the canonical 10/9, double-counting Uehling as a separate VP contribution).

The remaining −5.65 MHz residual is not attributed: the one-loop $\alpha(Z\alpha)^5$ term alone is about +6.9 MHz, and an earlier decomposition into multi-loop and non-loop pieces is withdrawn (Section 5.2). Sprint LS-7/LS-8a tested the two-loop leg and returned a WEAK verdict at the renormalization wall.

The structural reading is that the framework's bound-state QED machinery on Dirac-S^3 is built from three Paper 34 projections (Fock conformal + Camporesi–Higuchi spinor lift + Sturmian reparameterization), one new projection added during the sprint sequence (Drake–Swainson asymptotic subtraction, Paper 34 § III.13), and one standard textbook convention (Eides § 3.2 self-energy organization). No GeoVac-specific fitting parameter enters. The test of the framework here is its Bethe logarithms against the tabulated ones; the sub-percent headline itself is standard one-loop QED.

References

[1] GeoVac Project, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," Paper 7 (2025).

[2] GeoVac Project, "Structurally Sparse Qubit Hamiltonians from Natural-Geometry Hydrogenic Bases," Paper 14 (2025).

[3] GeoVac Project, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," Paper 18 (2026).

[4] GeoVac Project, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," Paper 28 (2026).

[5] GeoVac Project, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," Paper 34 (2026).

[6] GeoVac Project, "Time as Projection: The Klein-Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split," Paper 35 (2026).

[7] GeoVac Project, "LS-1 Sprint: Hydrogen Lamb Shift on the Dirac-S^3 Framework," `debug/ls1_lamb_shift_memo.md` (May 2026).

[8] GeoVac Project, "LS-2 Sprint: Bethe Logarithm from GeoVac Spectral Machinery," `debug/ls2_bethe_log_memo.md` (May 2026).

[9] GeoVac Project, "LS-3 Sprint: Regularized Bethe Logarithm via Acceleration Form," `debug/ls3_bethe_log_regularized_memo.md` (May 2026).

[10] GeoVac Project, "LS-4 Sprint: Bethe Logarithm via Schwartz / Drake–Swainson Regularization," `debug/ls4_bethe_log_drake_memo.md` (May 2026).

[11] GeoVac Project, "LS-5 Sprint: Two-loop QED Lamb shift on S^3 (scoping)," `debug/ls5_two_loop_scoping_memo.md` (May 2026).

[12] GeoVac Project, "LS-6a Sprint: Hydrogen Lamb Shift in the Eides § 3.2 Canonical Convention," `debug/ls6a_eides_convention_memo.md` (May 2026).

[13] H. A. Bethe, "The Electromagnetic Shift of Energy Levels," *Phys. Rev.* **72**, 339 (1947).

[14] A. Antognini *et al.*, "Proton structure from the measurement of 2S–2P transition frequencies of muonic hydrogen," *Science* **339**, 417–420 (2013).

[15] S. G. Karshenboim, "Precision physics of simple atoms: QED tests, nuclear structure and fundamental constants," *Phys. Rep.* **422**, 1–63 (2005).

[16] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, "CODATA recommended values of the fundamental physical constants: 2018," *Rev. Mod. Phys.* **93**, 025010 (2021).

[17] S. Navas *et al.* (Particle Data Group), "Review of particle physics," *Phys. Rev. D* **110**, 030001 (2024).

[18] M. I. Eides, H. Grotch, and V. A. Shelyuto, "Theory of light hydrogenic bound states," *Physics Reports* **342**, 63 (2001).

[19] P. J. Mohr, G. Plunien, and G. Soff, "QED corrections in heavy atoms," *Physics Reports* **293**, 227 (1998).

[20] C. Schwartz, "Lamb Shift in the Helium Atom," *Phys. Rev.* **123**, 1700 (1961).

[21] G. W. F. Drake and R. A. Swainson, "Bethe logarithms for hydrogen up to $n = 20$, and approximations for two-electron atoms," *Phys. Rev. A* **41**, 1243–1246 (1990).

[22] V. A. Yerokhin, Z. Harman, and C. H. Keitel, "Two-loop electron self-energy for low nuclear charges," Phys. Rev. Lett. **133**, 251803 (2024); arXiv:2411.12459.

[23] R. Camporesi and A. Higuchi, "On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces," *J. Geom. Phys.* **20**, 1 (1996).